

CHESAPEAKE LIGHT, USA

WMO: 994020

Lat: 36.910N

Lon: 75.710W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (W05)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.7	26.8	7.3	8.0	26.1	11.0	9.6	29.0	44.3	33.3	41.7	37.8	22.3	330	0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	84.4	78.3	82.6	78.0	81.3	77.6	81.6	82.7	80.3	81.5	79.2	80.4	14.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.3	162.7	82.0	80.1	156.1	80.9	78.9	149.9	79.9	45.2	82.6	43.9	81.6	42.7	80.5	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
49.4	40.9	35.4	18.3	88.5	5.8	1.7	14.1	89.7	10.7	90.7	7.5	91.6	3.2	92.9		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	41.3	41.7	47.1	55.4	63.3	72.5	77.1	76.7	72.5	64.3	54.5	47.5
	DBStd	14.16	8.71	7.34	6.97	6.38	5.83	4.97	3.01	3.06	3.73	5.62	6.86	7.42
	HDD50	870	290	247	140	22	0	0	0	0	1	35	135	
	HDD65	3288	733	654	554	294	103	8	0	1	80	318	543	
	CDD50	4373	22	14	51	183	412	676	841	828	675	445	170	56
	CDD65	1316	0	0	0	5	50	234	376	363	226	59	3	0
	CDH74	6597	0	0	0	5	75	919	2541	2340	666	51	0	0
	CDH80	630	0	0	0	0	5	91	291	213	29	1	0	0
Wind	WSAvg	18.1	19.6	20.0	21.4	21.0	18.4	15.5	14.5	14.3	16.8	17.5	18.9	19.1
Precipitation	PrecAvg	55.3	4.3	3.7	4.4	3.7	4.3	5.0	6.2	6.0	5.7	4.2	3.5	4.2
	PrecMax	73.7	7.5	7.8	8.8	7.0	8.0	9.4	10.0	11.5	19.9	8.7	9.3	8.6
	PrecMin	38.5	1.8	1.4	1.3	1.3	1.7	1.8	3.7	2.4	1.4	0.1	0.8	1.7
	PrecStd	7.4	1.4	1.6	1.5	1.4	1.7	1.9	1.5	2.1	3.9	1.9	2.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.6	63.4	68.3	73.4	79.3	85.7	87.5	87.0	82.9	78.2	70.0	66.5
		MCWB	59.5	58.2	58.8	62.8	70.6	78.0	79.7	79.6	77.1	73.5	69.1	64.0
	2%	DB	58.8	58.4	63.3	69.1	76.1	82.7	84.6	83.9	80.5	75.2	67.9	63.1
		MCWB	57.1	54.9	59.9	62.5	71.6	77.6	79.2	77.5	77.1	72.3	66.3	61.7
	5%	DB	56.1	54.5	59.8	66.7	73.3	80.3	82.9	82.1	79.1	73.7	65.7	60.5
		MCWB	54.2	50.8	55.7	61.7	70.2	76.4	78.7	77.8	76.2	71.5	63.4	59.4
	10%	DB	53.3	51.3	57.0	64.1	71.0	78.5	81.5	80.9	77.6	71.9	63.6	57.7
		MCWB	51.4	48.5	53.7	60.7	69.1	75.2	78.0	77.4	74.5	68.6	61.2	56.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	60.1	63.6	67.2	75.2	82.7	82.4	84.0	81.2	75.9	68.9	66.4
		MCDB	61.8	61.3	64.6	68.6	76.1	83.4	83.4	85.3	81.8	76.3	69.2	66.5
	2%	WB	58.3	56.0	60.7	64.9	73.1	80.1	81.5	81.3	79.4	74.2	66.2	63.3
		MCDB	59.1	57.6	62.7	66.7	74.3	81.2	83.1	81.7	79.9	74.3	66.6	63.9
	5%	WB	55.3	52.2	58.2	63.3	71.4	78.5	80.4	79.9	78.1	72.8	63.4	60.6
		MCDB	56.1	54.1	60.7	65.6	72.7	80.0	82.1	80.7	79.0	73.1	63.8	61.0
	10%	WB	52.1	48.8	54.8	61.3	70.0	77.1	79.3	78.8	76.5	70.2	61.4	58.2
		MCDB	53.4	50.6	56.7	63.8	71.4	79.0	80.9	79.8	77.6	71.2	62.1	58.8
Mean Daily Temperature Range	5% DB	MDBR	9.4	9.2	9.3	8.6	7.2	5.8	5.5	4.6	4.4	6.0	7.2	8.1
		MCDBR	12.0	14.1	14.0	12.3	10.8	8.6	7.8	6.9	6.2	7.4	8.5	10.7
		MCWBR	11.8	13.3	11.1	9.0	7.2	6.4	6.3	6.2	6.0	7.7	9.5	10.9
	5% WB	MCDBR	11.1	14.4	14.1	11.1	8.1	8.0	7.1	5.9	5.6	6.1	7.7	10.3
		MCWBR	11.9	14.4	12.5	9.8	6.8	6.9	6.2	6.3	5.7	7.2	9.7	11.1
Clear-Sky Solar Irradiance	taub	0.316	0.325	0.349	0.396	0.441	0.485	0.514	0.506	0.414	0.360	0.329	0.319	
	taud	2.547	2.516	2.478	2.359	2.267	2.194	2.123	2.132	2.388	2.529	2.558	2.564	
	Ebn at Noon	279	290	291	281	269	256	247	246	266	273	272	270	
	Edh at Noon	24	29	32	38	43	46	49	48	35	28	24	23	
All-Sky Solar Radiation	RadAvg	723	975	1322	1690	1908	2048	1957	1766	1448	1134	822	648	
	RadStd	58	97	101	115	158	130	93	110	109	113	64	51	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (50 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+295	+187

Nomenclature: See separate page